QUANTITATIVE RESEARCH

Game-Theoretic Frameworks for Crypto Perpetual Trading Strategies

Strategic Microstructure Analysis & Execution Playbooks

AUTHOR

Taha Fadil Rezok

DATE

May 24, 2025

CONTENTS

STRATEGY 01 / 07 · STACKELBERG GAME AGAINST INFORMED FLOW

Adaptive Guerrilla Market Making

Avellaneda-Stoikov reservation pricing combined with real-time toxic flow deflection

MM CORE **ADAPTIVE_GUERRILLA.py** Continuous Stackelberg · Hit-and-Run Quoting

UNDERLYING GAME

Market making is a continuous Stackelberg game against informed flow. The classic Avellaneda-Stoikov model yields the optimal reservation price and spread as a function of inventory, time-to-liquidation, and realized volatility. Guerrilla tactics add a second layer: we post aggressively in the direction we want the market to believe — to deflect toxic flow — then cancel/repost instantly when `VPIN` or `OFI` turns against us. This is "hit-and-run" quoting: **high cancellation rate, low adverse selection**.

> **Dominant Strategy:** Inventory skew (Avellaneda-Stoikov) + toxic flow deflection (guerrilla) creates a dominant strategy against the average MM. The equilibrium: informed traders hit our quotes when too tight → we detect toxicity faster and cancel → uninformed flow fills us when correctly skewed.

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Informed Takers	Extract α from mispriced quotes	Hit our quotes when mispriced → we cancel. Price-sensitive, directional flow.
Uninformed Seekers	Execute at best available price	Provide our spread capture. No directional signal. Source of clean PnL.
Other MMs	Quote competitively, earn spread	We front-run their quotes by being faster on cancellation and re-entry.
Us	**Maximize spread capture, minimize adverse selection**	Optimal reservation price + real-time toxicity filter (VPIN + OFI). Cancel before the adverse fill lands.

INVALIDATION CONDITIONS

- × Regime change to **extreme illiquidity** — spread multiplier >4× baseline; hit-and-run quoting becomes adversely selected even on fast cancel.
- × Inventory becomes **extremely one-sided** without a mean-reversion signal; skew fails to attract offsetting flow and compounds directional risk.
- × Venue switches to **pro-rata matching** — FIFO queue value collapses, cancellation-priority edge is fully invalidated.

STRATEGY 02 / 07 · CONVERGENCE GAME WITH TIMING EXTERNALITIES

Funding Rate Arbitrage

Exploiting the crowded-trade unwind around perpetual funding settlement clocks

ARB / CARRY **FUNDING_ARBITRAGE.py** Timing Game · Prisoner's Dilemma Unwind

UNDERLYING GAME

Perpetual funding rates create a repeated convergence game with a hard **8-hour settlement clock**. When funding is extreme (e.g., +0.1% per 8h ≈ 45% annualized), crowded longs are paying heavily to maintain positions. The trade (short perp + long spot) is known to all sophisticated participants, creating a coordination problem: everyone wants to enter before the print but exit just after.

→ **Entering too early:** suffer adverse perp price movement before the print.

→ **Entering too late:** miss the alpha — others have front-run the entry window.

→ **Optimal entry:** when `VPIN` of the spot/perp divergence exceeds threshold AND print is within the entry window.

> **Nash Outcome — Prisoner's Dilemma of the Unwind:** If all arbs unwind simultaneously after the print, they create a buy cascade (short covering on perp) which moves the perp premium against them. Everyone races to unwind first → price reverts sharply right after the funding print → **predictable 5-minute pattern** we systematically exploit.

SUB-STRATEGY A · FUNDING CAPTURE

- Long-hold delta-neutral: short perp / long spot
- Collect funding for multiple 8h epochs
- Risk: trend continuation against the carry leg
- Exit signal: funding normalization or OI collapse

SUB-STRATEGY B · PRE-PRINT REVERSAL

- Enter counter-position 30–60 min before print
- Condition: rate confirmed extreme + Spot CVD diverging
- Exit within 5 min of funding print
- Pure gamma play on the mechanical unwind

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Arbitrageurs (us + peers)	Capture funding / reversion alpha	Short perp, long spot. Rational agents modeling each other's behavior and entry timing.
Retail Longs	Momentum / long-term hold	Paying funding due to behavioral bias. Cash flow source we collect from.
Exchange	Balance open interest	Designs funding formula to discourage extremes; rates persist via retail bias regardless.
Us	**Capture carry + gamma on unwind**	VPIN-timed entry. Monitor unwind race dynamics; exit before crowd reversal compresses edge.

INVALIDATION CONDITIONS

× Spot CVD confirms the perp direction — genuine buying on spot means no distribution; retail is right, do not fade.

× OI expanding *into* the print — new longs still entering implies rate may go higher before reverting.

× Macro catalyst within the print window (FOMC, CPI) — fundamental flow overrides mechanical convergence.

× Basis widens further after entry — equilibrium level estimate is wrong; cut immediately.

STRATEGY 03 / 07 · BAYESIAN UPDATING GAME

Glosten-Milgrom Adverse Selection Model

Information asymmetry quantification via VPIN and Kyle's Lambda for dynamic spread calibration

RISK MODEL **INFO_ASYMMETRY.py** Glosten-Milgrom · Bayesian Belief Updating

UNDERLYING GAME

The market maker faces a binary-type population: **informed traders** (who know the true value V*) and **uninformed noise traders** (who trade for liquidity reasons). The MM sets a spread wide enough to break even across both types. In equilibrium: `spread = f(π, V_range)` where `π` is the informed fraction.

KEY QUANTITATIVE MEASURES

$$VPIN = |V_{buy} - V_{sell}| / V_{total}$$

Volume-Synchronized Probability of Informed Trading — empirical real-time estimator of π (informed fraction)

$$\Delta P = \lambda \cdot OFI + \varepsilon$$

`λ (Kyle)`	Market depth / impact coefficient. High λ → each unit of OFI moves price more → toxic informed flow. Estimated via rolling OLS on (ΔP, OFI) pairs.
`OFI`	Order Flow Imbalance: net signed aggressive volume in the measurement window
`ε`	Residual noise component (uninformed flow contribution)

DECISION PROTOCOL ON VPIN SPIKE

1. As a **market maker:** widen spreads or pause quoting immediately — do not get adversely selected into an informed fill.
2. As a **directional trader:** informed flow provides signal — fade the uninformed side, follow the informed side.

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Informed Trader	Extract α from private information V*	Knows V* > current price. Buys aggressively. Appears as persistent buy CVD without corresponding sell volume.
Uninformed MM (victim)	Earn spread	Posts tight spreads; gets adversely selected. Loses money on every informed fill. Cannot distinguish ex-ante.
Us	**Monitor and exploit information signals**	Monitor VPIN and λ. VPIN spike → widen / pull quotes. Directional signal extractable → join the informed side.

INVALIDATION CONDITIONS

- × VPIN spike driven by a **mechanical hedger** — check OI: if OI is flat while volume spikes, it is a hedger, not a speculator. No directional signal.
- × `λ` is high but CVD is oscillating — this is noise, not signal; do not enter a directional position.
- × Funding extremes attracting **pure arbitrage flow** — not price-discovery flow; Glosten-Milgrom model inapplicable in this regime.

STRATEGY 04 / 07 · DOMINATED STRATEGY EXPLOITATION

Liquidation Engine Front-Running

Exploiting the zero-optionality of forced liquidation engines via cascade mechanics

---

PREDATORY **LIQUIDATION_FRONTRUN.py** Dominated Strategy · Cascade Alpha

UNDERLYING GAME

A liquidation engine is an algorithmic player with a **single dominated strategy**: when a position's margin falls below maintenance threshold, it *must* sell (or buy) at market at any price until the position is closed. The liquidation engine has no ability to deviate, delay, or bluff. **The game becomes one-sided: we know the engine's action before it takes it; it doesn't know ours.**

CASCADE MECHANICS — THE REAL PRIZE

- → **Step 1:** Price drops X% → 10× leverage positions liquidated at market. This sell pushes price further into the next leverage band.
- → **Step 2:** 8× leverage positions hit. Each liquidation creates the price move that triggers the next — self-reinforcing loop.
- → **Entry discipline:** enter at the START of the cascade, exit into the forced selling flow.

SECOND TRADE — ABSORPTION ALPHA

- → **Exhaustion signal:** volume spike without further price movement — the absorption candle. No more leveraged positions in the band.
- → **Absorption trade:** place limit bids deep below market; capture the reversion as forced selling dries up entirely.

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Liquidation Engine	Close position at any price	Completely predictable. Must sell/buy regardless of price. Zero optionality. Our most reliable counterparty.
Leveraged Retail	Hold leveraged position	10–20× positions with stops clustered predictably (below round numbers, below recent lows). Cascade fuel.
Us — Cascade Entry	**Capture directional cascade alpha**	Enter at cascade start, exit passively into forced selling. High-conviction, time-bounded.
Us — Absorption MM	**Capture post-exhaustion reversion**	Deep limit orders below cascade start. Collect from overshoot. Profit from reversion as liquidations dry up.

> **INVALIDATION CONDITIONS**
>
> - × OI is **declining before cascade start** — positions already deleveraged; fewer stops to hit → cascade will not propagate to next band.
> - × CVD confirms cascade direction *after* the cascade — real sellers, not force-liquidated ones. This is **trend continuation, not reversion**. Do not fade.
> - × Exchange halt or **socialized loss event** — catastrophic for absorption trade; cascade may not revert if insurance fund is depleted.
> - × Funding rate does not normalize post-cascade — still crowded the other way; mean-reversion thesis is incomplete.

Predatory Liquidity — Stop Hunt Engineering

Self-fulfilling coordination equilibria around resting stop clusters

DIRECTIONAL **PREDATORY_LIQUIDITY.py** Coordination Equilibrium · Stackelberg Leadership

UNDERLYING GAME

Stop hunts are **coordination equilibria**. Multiple informed participants independently identify the same liquidity pool — equal highs with resting stop-longs, equal lows with resting stop-shorts. Because reaching the liquidity cluster pays off only if enough volume moves price there, each predator's action is justified *because* others will also act: a **self-fulfilling prophecy of the sweep**.

Stackelberg Leader: The participant who moves first at scale forces price through the level, triggering stops and creating the very volume that confirms the move to followers. Followers' taker orders amplify the leader's impact — a **positive-feedback cascade**. Leader captures the full sweep; follower pays slippage into it.

STRATEGY A · JOIN THE HUNT

- Enter in sweep direction before cascade
- Exit passively once stop cluster is hit
- Must lead — not follow. Latency-critical.
- Abort immediately if OI expands through level

STRATEGY B · FADE POST-SWEEP

- After stop liquidity is fully exhausted
- Signal: absorption candle + vol spike + CVD reversal
- Enter against sweep direction — real price reasserts
- Stop-hunt premium collapses to fair value

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Predators (us + similar)	Capture stop-cluster liquidity alpha	Identify equal highs/lows and OI-confirmed stop clusters. Enter early; exit passively into triggered-stop liquidity.
Victims (Retail)	Hold technical positions	Stops at predictable levels — below equal lows, above equal highs. Created the very target predators aim for.
Passive MMs	Provide liquidity at fair value	Retreat as vol spikes during the sweep. Spread widening accelerates the move — inadvertent enablers.
Us	**Lead or fade the stop hunt**	Stackelberg leader role on high-conviction setups. Post-sweep fader when absorption confirmed via CVD reversal.

INVALIDATION CONDITIONS

- × Fundamental news driving **genuine directional flow** — stops hit for a legitimate reason, not manipulation. Check macro calendar first.
- × OI **expanding through the sweep level** — new positions opening, not stops being hit. Genuine breakout; abort the fade entirely.
- × CVD remains in sweep direction after the spike — informed flow continues; sweep has become a directional trend.
- × Price fails to retrace >30% of the sweep within 5 minutes — no absorption confirmed; the hunt has not exhausted.

STRATEGY 06 / 07 · STACKELBERG QUEUE LEADERSHIP + FIFO OPTIONALITY

Queue Warfare — FIFO Priority Exploitation

Optimal queue management under time-priority matching: when to join, hold, and cancel

EXECUTION **QUEUE_WARFARE.py** Stackelberg Within Price Level · FIFO Optionality

UNDERLYING GAME

Order books with FIFO matching create a **time-priority auction**. Being first in queue at a price level is a valuable option: you fill before anyone else at that price if the level trades. This option value is proportional to the probability of fill and inversely proportional to the adverse selection cost per fill.

Cancel when: P(fill | informed) > P(fill | noise) × E[profit per fill]

When OFI shifts against your quote, the imminent fill is more likely adverse than profitable. Cancel before it lands.

THE STRATEGIC TENSION

- → **Joining early** → higher fill probability, but more time exposure to adverse selection (queue fills because informed flow arrives).
- → **Joining late** → lower fill probability and lower adverse selection cost per fill — systematically worse EV for noise capture.
- → **Cancelling and re-entering** → depletes queue priority; the cost is the surrendered FIFO position — irreversible.

This is a Stackelberg game within a single price level: the **Leader** holds first-fill optionality but bears first adverse selection risk — the exact tradeoff this strategy manages dynamically.

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Queue Leader (us, ideally)	First fill on noise; first cancel on toxic	Earliest order placement. First in line. Highest fill prob on noise. Must cancel fastest on toxic detection.
Queue Follower	Fill on residual noise flow	Joined late. Fills after us. Lower adverse selection but lower capture rate — systematically worse EV.
Iceberg Orders	Hide true order size	Hidden reserve at a level; consume queue space without revealing depth. Detecting icebergs critical for fill prob model.
Market Order Flow	Execute immediately	Both the "prize" (uninformed fill) and the "threat" (informed taker draining queue adversely).
Us	**Maximize P(fill | noise) × profit**	Maintain queue leadership on FIFO venues. Cancel on OFI signal before adverse fill lands. Detect iceberg levels to calibrate fill probability accurately.

INVALIDATION CONDITIONS

- × Venue switches from **FIFO to pro-rata matching** — queue position loses all value; cancel logic is irrelevant. Strategy completely inapplicable.
- × **Latency disadvantage** — 2nd or 3rd in queue means cancel logic is moot; adverse selection hits regardless. Co-location is mandatory.
- × Queue is **icebergd** — visible size is a fraction of real depth → fill probability model is systematically wrong; do not rely on visible book.

STRATEGY 07 / 07 · CRAWFORD-SOBEL SIGNALING GAME

Spoof Detection & Counter-Positioning

Detecting and exploiting cheap-talk order book manipulation in perpetual markets

COUNTER-INTEL `SPOOFING_COUNTER.py` Crawford-Sobel · Cheap-Talk Signaling

UNDERLYING GAME

A spoofer plays a **cheap-talk signaling game**. They place large visible orders with zero genuine intent to fill — the signal is costless because cancellation is free. Uninformed participants update beliefs about true supply/demand from this false signal, adjusting quotes and taking positions that favor the spoofer's real objective.

> **Crawford-Sobel Analysis:** Cheap talk is informative only when sender incentives are partially aligned with the receiver's. Here they are not — the spoofer profits *specifically* when the receiver's beliefs are wrong. In the limit with perfect detection, the equilibrium collapses: spoofing has zero EV if all receivers ignore it. **Our edge lives in the delta between the market's imperfect detection and ours.**

SPOOF MECHANICS — THE MANUFACTURED MOVE

- → Spoofer places a **large visible bid** to attract buy-side FOMO and tighten ask-side quotes from rule-based MMs.
- → Uninformed participants react: tighten asks or delay sell orders — exactly what the spoofer needs to execute.
- → Spoofer **sells into the manufactured demand**, then cancels the bid. Price reverts as false support disappears.

DETECTION SIGNALS

- → **CVD divergence:** large bid visible but buy-side CVD does not confirm — no genuine aggressive buying behind the signal.
- → **Zero partial fills:** a real large buyer would partially fill as price approaches; a spoofer cancels clean without any fills.
- → **Cancel timing:** spoof orders cancel as soon as the desired price movement is achieved — detectable via order lifecycle analysis.

PLAYER PROFILES

PLAYER	OBJECTIVE	BEHAVIOR & INTERACTION
Spoofer	Manufacture price movement to exit favorably	Large visible order, zero fill intent. Sells into FOMO demand, cancels bid. Often a directional trader or slow MM unwinding.
Victim (Retail Algos)	React to book depth as direction signal	Uses raw book depth as direction indicator. Tightens ask when large bid appears — exactly what the spoofer needs.
Us	**Detect spoof, fade false signal, capture reversion**	Detect via CVD/fill-rate divergence. Fade the false price signal. Position for mean-reversion once the spoof order is pulled.

INVALIDATION CONDITIONS — STOP PLAYING THE FADE

- × CVD **corroborates the spoof direction** — genuine aggressive flow at that level; this is real demand, not a spoof. Abort the fade immediately.
- × A large **block taker order hits the "spoof" level** — someone is absorbing, not spoofing. Level has real support; do not fade.
- × Realized volatility spikes above **2σ** — regime change; false signals proliferate; spoof detection model produces systematic false positives.
- × Market-wide liquidation cascade underway — order book is structurally unreliable; detection framework inapplicable.